m1OASYS ASCOM Driver

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Introduction

ASCOM dome/roof driver for observatories using the ELK M1 Gold and m1OASYS architecture.

Overview

Introduction

Welcome to m1OASYS

m1OASYS is an ASCOM-compliant observatory roof driver designed specifically for observatories controlled by an Elk M1 Gold security and automation system. The driver provides a reliable interface between ASCOM-compatible astronomy applications and the observatory roof, allowing automated roof operation while maintaining multiple layers of operational safety.

The driver supports roll-off roofs, clam-shell roofs, and other observatory enclosures that can be controlled through Elk M1 outputs and monitored through Elk M1 zones. By leveraging the Elk M1's proven reliability and extensive I/O capabilities, m1OASYS provides a robust observatory automation solution suitable for both attended and unattended operation.

Key Features

ASCOM Dome Interface

m1OASYS implements the ASCOM Dome interface, allowing compatibility with a wide range of astronomy applications including:

- NINA
- Voyager
- ACP
- TheSkyX
- Sequence Generator Pro
- ASCOM Remote / Alpaca
- Other ASCOM-compliant software

Roof Control

The driver supports:

- Open roof commands
- Close roof commands
- Abort roof movement
- Roof status reporting
- Open and closed limit monitoring

Mount Safety Integration

Optional mount safety monitoring prevents unsafe roof movement when a telescope is not in a known safe position.

Features include:

- Safe-to-open verification

- Safe-to-close verification
- Continuous safety monitoring
- Configurable enable/disable operation

Pulse Telemetry

For installations equipped with a pulse or hall-effect sensor, m1OASYS can provide advanced roof position tracking.

Features include:

- Real-time pulse counting
- Roof percentage open calculation
- Automatic calibration
- Motion verification
- Position reporting

Motion Verification

Optional motion verification enhances operational safety by confirming that roof movement is actually occurring after a movement command is issued.

Features include:

- Motion timeout detection
- Stalled roof detection
- Fault reporting
- Automatic safety shutdown

Fault Detection and Recovery

m1OASYS continuously monitors roof operation and can detect a variety of abnormal conditions.

Examples include:

- Motion timeout faults
- Communication failures
- Safety violations
- Position inconsistencies

The driver maintains detailed fault information and provides diagnostic information through the Status Window.

Real-Time Status Window

An integrated status window provides live telemetry and operational information.

Displayed information includes:

- Roof state
- Percent open
- Pulse count
- Mount safety status
- Fault status
- Last fault time
- Reconnect history

- Watchdog activity

Multi-Controller Support

Current versions support multiple controller architectures including:

- Single-button motor controllers
- Dedicated Open/Close controllers
- Dedicated Open/Close/Stop controllers

This allows the driver to interface with a wide variety of commercial and custom roof control systems.

ASCOM Remote and Alpaca Compatibility

m1OASYS can be used through ASCOM Remote Server, making the observatory available over an Alpaca network connection.

This allows remote control from compatible Alpaca clients while preserving the full functionality of the native ASCOM driver.

Design Goals

m1OASYS was designed with the following objectives:

- Reliability
- Safety
- Simplicity
- Compatibility
- Fault tolerance
- Observatory automation

The driver emphasizes predictable operation and safe recovery from abnormal conditions while remaining easy to configure and maintain.

Intended Audience

This documentation is intended for:

- Observatory owners
- Astrophotographers
- Remote observatory operators
- Observatory system integrators
- ASCOM software users
- Elk M1 automation users

Whether operating a backyard roll-off roof observatory or a fully automated remote installation, m1OASYS provides a flexible and reliable roof automation platform integrated with the ASCOM ecosystem.

No warranty as to the operation or automation of the observatory equipment; it simply provides components for ‘Do It Yourself’ automation projects.

Caution! Automation of any rolling roof observatory requires special attention with regard to the location of individuals in the observatory, as well as the position of the telescope/mount, prior to moving the roof to ensure no one or nothing obstructs the roofs path. Serious injury to individuals and/or damage to telescope, mount or the observatory may occur if this advice is not followed. We recommend that all users of m1 ASCOM driverLive implement safety policies and procedures regarding the operation of their rolling roof. In addition we strongly recommend installation of safety sensors that will prevent or stop movement of the roof should individuals or equipment move into an unsafe position before or during roof movement. For remote/automated operation of rolling roofs it is recommended that the "*Allow Remote Apps to Open/Close the Roof*"' check-box in the m1 ASCOM driverLive Live software be unchecked if the roof does not safely clear the scopes/mounts. Misuse or careless operation can cause serious damage to your equipment or individuals. The author of the software and the distributor of the equipment (1) make no warranty or representation to the user as to the operation of this System and (2) is NOT responsible for any injury or damage that may occur from careless implementation, operation or misuse. **Opening or Closing a roll-off roof is not a trivial task and the operator needs to take all necessary precautions before starting the roof motor. Damage to equipment or persons can occur when the roof is in motion so all necessary precautions should be taken before executing such a command.**

System Requirements

Hardware Requirements

- Windows PC
- ASCOM Platform
- ELK M1 Gold with XEP Ethernet interface
- Configured ElkRP automation rules
- Optional Hall-effect pulse sensor
- Optional scope-safe sensors
-

Driver Architecture

Component	Responsibility
ELK M1	Hardware authority and sensor evaluation
ASCOM Driver	Roof control, telemetry, safety enforcement
Client Software	Observatory sequencing and telescope control

Elk m1 Rules, Texts, Counters, Etc.

ELK M1 Configuration

Overview

The m1OASYS ASCOM Roof Driver communicates directly with the ELK M1/XEP using Elk ASCII text commands over TCP/IP.

The ELK provides:

- Hardware relay control
- Roof position sensing
- Optional Hall pulse telemetry
- Optional scope-safe interlock evaluation
- Status responses back to the ASCOM driver

The ASCOM driver is responsible for:

- Roof automation
- Telemetry interpretation
- Safety enforcement
- Calibration
- Watchdog protection
- ASCOM interoperability

Required ELK Configuration

Required Tasks

The following ELK tasks are expected by the driver.

Task	Purpose
Task 1	Open Roof
Task 2	Close Roof
Task 3	Stop Roof

Example names:

Task Number	Suggested Name
1	Open Roof
2	Close Roof
3	Stop Roof

Required Zones

Roof Open Limit

Zone	Purpose
Zone 11	Roof Opened Sensor

Expected behavior:

- NOT SECURE = Roof fully open

Roof Closed Limit

Zone	Purpose
Zone 12	Roof Closed Sensor

Expected behavior:

- SECURE = Roof fully closed

Optional Zones

Scope Safety Sensor

Zone	Purpose
Zone 7	Scope Safe Sensor

Expected behavior:

- SECURE = Telescope safe for roof motion
- NOT SECURE = Unsafe for roof motion

This zone is only required if:

- "Enable Mount Safety Sensors" is enabled in the ASCOM Setup Dialog.

Hall Pulse Sensor

Zone	Purpose
Zone 5	Roof Motion Pulse Sensor

Used for:

- pulse counting
- calibration
- percent-open tracking
- watchdog protection

This zone is optional.

Required Counters

Roof Pulse Counter

Counter	Purpose
Counter 7	Roof motion pulse count

The driver polls this counter during motion when pulse telemetry is enabled.

Required Text Definitions

Open Roof Command

Text Number	Value
2	09tn00100C4^M

Close Roof Command

Text Number	Value
3	09tn00200C3^M

Roof Status Query

Text Number	Value
7	09xx00100B6^M

Scope Safety Query

Text Number	Value
20	11xx005sensoron0042^M

Pulse Counter Query

Text Number	Value
25	07cv00729^M

Required Driver Responses

Roof Open Response

ELK sends:

0DXX001open0039^M

Driver interprets:

- roof fully open

Roof Closed Response

ELK sends:

0FXX001closed006F^M

Driver interprets:

- roof fully closed

Roof Unknown Response

ELK sends:

10XX001unknown00EE^M

Driver interprets:

- roof moving / intermediate state

Scope Safety Responses

Scope Safe

ELK sends:

0FXX002Secure0081^M

Driver interprets:

- roof motion allowed

Scope Unsafe

ELK sends:

12XX002NotSecure0063^M

Driver interprets:

- roof motion blocked

Required Automation Rules

Open Roof Task

WHENEVER Open Roof (Task 1) IS ACTIVATED

THEN TURN Open Roof Relay ON

Close Roof Task

WHENEVER Close Roof (Task 2) IS ACTIVATED

THEN TURN Close Roof Relay ON

Stop Roof Task

WHENEVER Stop Roof (Task 3) IS ACTIVATED

THEN TURN Open Roof Relay OFF

THEN TURN Close Roof Relay OFF

Roof Status Query Rules

The driver periodically requests roof status using:

09xx00100B6^M

ELK must respond with:

- OPEN
- CLOSED
- or UNKNOWN

based on current limit switch state.

Pulse Counting Rule

WHENEVER Roof Pulse Zone BECOMES NOT SECURE

THEN ADD 1 TO Counter 7

Pulse Counter Reset Rules

WHENEVER Open Roof Task ACTIVATES

THEN SET Counter 7 TO 0

WHENEVER Roof Closed Sensor BECOMES SECURE

THEN SET Counter 7 TO 0

Scope Safety Rules

WHENEVER THE FOLLOWING TEXT IS RECEIVED:

11xx005sensoron0042^M

AND Scope Safe Zone IS SECURE

THEN SEND:

0FXX002Secure0081^M

WHENEVER THE FOLLOWING TEXT IS RECEIVED:

11xx005sensoron0042^M

AND Scope Safe Zone IS NOT SECURE

THEN SEND:

12XX002NotSecure0063^M

Notes

CRC Framing

The driver internally generates CRC-framed Elk packets automatically.

ELK rules must match the fully framed transmitted text including:

- packet length
- CRC
- carriage return (^M)

Optional Features

The following driver features are optional:

Feature	Required Hardware
Pulse Telemetry	Hall sensor
Percent Open Tracking	Hall sensor
Scope Safety Interlock	Scope-safe sensor

The driver supports traditional limit-switch-only operation when optional hardware is not installed.

Recommended Architecture

ELK Responsibilities

- Relay control
- Sensor authority
- Pulse counting
- Safety state determination

ASCOM Driver Responsibilities

- Roof automation
- Telemetry processing
- Calibration
- Watchdog protection
- Safety enforcement

Client Software Responsibilities

- Observatory sequencing
- Telescope parking
- Imaging automation

This separation provides a reliable and maintainable observatory control system.

Optional

Installation

m1 ASCOM driver is available via Internet download.

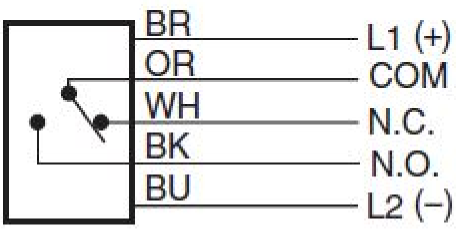
Important: m1 ASCOM driver requiresthe installation of the ASCOM Platform, available at http://ascom-standards.org.

Internet Installation

1. Download the latest m1 ASCOM driver driver and software.
2. You will be prompted to either ‘Run’ the installer or you can save it to a folder; double clicked the saved file with start the installer.
3. Follow the on screen instructions to complete the installation.
4. Once the software has successfully installed you must enter the setup areas to input the proper setting before attempting to connect your hardware.

Scope Position Sensor

Eaton 1451E



Supply power to the sensor via the m1 ASCOM driver board’s +12v (brown) and –12v (Blue), then connected the common (Orange) and normally open (N.O. Black) sensor wires to the appropriate zone input.

Create rules to ensure the sensor is secure before moving the roof, and you can even wire the power through a m1 ASCOM driver output relays to give you control of turning the IR beam on and off before and after opening and closing commands are sent.

Rule examples:

WHENEVER THE FOLLOWING TEXT IS RECEIVED: "11xx005sensoron0042^M" THROUGH PORT 0
THEN TURN Scope IR Sen (Out 13) ON

WHENEVER Open Roof (Task 1) IS ACTIVATED
AND Scope 1 Sensor 1 (Zn 7) IS SECURE
THEN TURN Open Roof (Out 9) ON FOR 5 SECS

Setup

Setup Configuration

1. Install the ASCOM Platform.
2. Install and register the m1OASYS ASCOM Roof Driver.
3. Configure the ELK M1 XEP IP address and port in the Setup Dialog.
4. Enable optional pulse telemetry if a Hall sensor is installed.
5. Enable optional scope safety interlock if safety sensors are installed.
6. Verify ElkRP text and automation rules are configured properly.
7. Connect the driver from NINA or another ASCOM client.

Hall Sensor Setup

Hall Pulse Counting Configuration

Overview

The m1OASYS Roof Driver supports optional Hall Effect pulse telemetry for accurate roof position tracking and motion verification.

A Hall Effect sensor mounted on the roof drive mechanism generates pulses as the roof moves. The Elk M1 counts these pulses and reports the accumulated count to the ASCOM driver.

The driver uses the pulse count to determine:

- Roof position percentage
- Roof movement verification
- Stall detection
- Calibration values
- Open and closed position tracking

Hardware Requirements

Required components:

- Hall Effect sensor
- Magnet mounted to roof drive gear, wheel, or shaft
- Elk M1 zone input
- Elk M1 counter

Each time the magnet passes the Hall sensor, a pulse is generated.

Elk M1 Zone Configuration

The Hall sensor should be connected to an unused Elk M1 zone.

Example:

```
Zone 5 = Roof Pulse Sensor
```

Recommended Elk configuration:

Setting	Value
Definition	16 - Non Alarm
Area	1
Type	Normally Open or Normally Closed (depending on sensor wiring)
Description	Roof Pulse

The zone is used strictly for pulse counting and should not generate alarm conditions.

Elk Counter Configuration

Create an Elk counter to store accumulated roof pulses.

Example:

```
Counter 7 = Roof Pulse Count
```

The driver requests the current counter value and uses it as the live pulse count.

Elk Rule Configuration

Create a rule that increments the pulse counter whenever the Hall sensor changes state.

Example:

```
Whenever Zone 5 Becomes Not Secure
    Then Increment Counter 7
```

Depending on the Hall sensor wiring and pulse frequency, either Secure or Not Secure transitions may be used.

Verify that the counter increases as the roof moves.

Driver Configuration

Enable pulse telemetry in the m1OASYS Setup Dialog.

Configuration options include:

- Enable Pulse Telemetry
- Open Pulse Count
- Motion Verification

The Open Pulse Count represents the number of pulses required for a complete roof travel from fully closed to fully open.

Calibration Procedure

1. Close the roof completely.
2. Reset the Elk pulse counter to zero.

3. Enable calibration mode.
4. Open the roof completely.
5. Record the final pulse count.
6. Enter the value as the Open Pulse Count.

Example:

```
Counter 7 Final Value = 5124

Open Pulse Count = 5124
```

The driver will automatically calculate:

```
Percent Open =
(Current Pulse Count ÷ Open Pulse Count) × 100
```

Motion Verification

When pulse telemetry is enabled, the driver continuously verifies roof movement.

If roof motion is commanded but the pulse count does not change, the driver assumes movement has stopped unexpectedly.

The driver will:

- Stop roof movement
- Enter fault state
- Report an error to ASCOM clients
- Generate a Roof Fault notification if enabled

This protects against:

- Broken drive chains
- Failed motors
- Slipping couplers
- Mechanical jams
- Failed Hall sensors

Telemetry Display

The Diagnostics Window displays:

- Current Pulse Count
- Open Pulse Count
- Percent Open
- Motion State
- Roof State
- Fault Status

Values update automatically while the roof is moving.

Troubleshooting

Pulse Count Does Not Change

Verify:

- Hall sensor power
- Sensor wiring
- Magnet placement
- Elk zone programming
- Elk counter rule operation

Use ElkRP to confirm that the counter increases when the magnet passes the sensor.

Roof Reports No Pulse Movement

Possible causes:

- Hall sensor failure
- Broken wiring
- Magnet missing
- Excessive magnet gap
- Mechanical drive failure

The driver will enter a fault state if movement is commanded but no pulse activity is detected.

Percent Open Incorrect

Verify:

- Open Pulse Count calibration value
- Elk counter reset before calibration
- Hall sensor pulse reliability
- Magnet placement consistency

Recalibrate if roof mechanics have changed.

Recommended Installation

For best accuracy:

- Mount the magnet on the main roof drive shaft or gear.
- Use a single magnet for consistent pulse spacing.
- Ensure the Hall sensor produces clean transitions.
- Position the sensor to avoid vibration or intermittent triggering.

A properly calibrated Hall telemetry system provides highly accurate roof position reporting and additional protection against mechanical failures.

Hall Sensor Calibration

Calibration

The driver supports automatic roof calibration using Hall-effect pulse counting. During calibration, the roof opens fully while the driver records the pulse count. The learned value is saved automatically and used for percent-open calculations.

Pushover Notifications

Pushover Notifications

m1OASYS can send real-time notifications to a mobile device using the Pushover notification service. Notifications are delivered directly from the driver and can be customized to report only the events that are important to the user.

Requirements

To use Pushover notifications:

1. Create a Pushover account at https://pushover.net.
2. Create a new Pushover application (for example, "m1OASYS").
3. Copy the application's **API Token/Key**.
4. Copy your personal **User Key** from the Pushover dashboard.
5. Open the m1OASYS Setup dialog.
6. Enable **Pushover Notifications**.
7. Enter the Application Token and User Key.
8. Use the **Test Notification** button to verify operation.

Notification Settings

Individual notification types can be enabled or disabled using the Setup dialog.

Available notification categories include:

- Roof Opened
- Roof Closed
- Roof Fault
- Communication Lost
- Communication Restored
- Scope Unsafe Blocked Movement

Roof Opened

Sent when the observatory roof reaches the fully open position.

Example:

□□Observatory roof opened

Roof Closed

Sent when the observatory roof reaches the fully closed position.

Example:

□□Observatory roof closed

Roof Fault

Sent when a roof fault condition is detected.

Examples:

□Roof fault: No pulse movement

□Roof fault: Movement timeout

Scope Unsafe Blocked Movement

Sent when m1OASYS prevents roof movement because the telescope is not in a safe position.

Example:

□Roof movement blocked - telescope not safe

Communication Lost

Sent when m1OASYS detects loss of communication with the ELK M1 controller.

Example:

□ELK communication lost

Communication Restored

Sent when communication with the ELK M1 controller is restored.

Example:

□ELK communication restored

Notes

- Notification settings are stored in the ASCOM Profile and persist between sessions.
- Each user supplies their own Pushover Application Token and User Key.
- Credentials are not included with the driver and are not stored in source code repositories.
- Notifications can be enabled individually to reduce unnecessary alerts.
- Communication monitoring uses telemetry activity to detect communication failures, including cases where the network connection remains open but valid controller responses stop.

Features

Features

- ASCOM Dome interface support
- Open / Close / Abort roof control
- Live shutter state telemetry
- ELK M1 TCP/IP integration
- Optional Hall-effect pulse telemetry
- Automatic roof calibration
- Percent-open roof tracking
- Movement timeout watchdog
- Pulse watchdog protection
- Optional scope-safe interlock support
- Persistent calibration storage
- Integrated live telemetry status window

Supported Software

Supported Applications

- NINA
- Voyager
- Sequence Generator Pro (SGP)
- CCD Commander
- Other ASCOM automation software

Hardware Required

Hardware Requirements

- Windows PC
- ASCOM Platform
- ELK M1 Gold with XEP Ethernet interface
- Configured ElkRP automation rules
- Optional Hall-effect pulse sensor
- Optional scope-safe sensors

Setup

Setup Configuration

1. Install the ASCOM Platform.
2. Install and register the m1OASYS ASCOM Roof Driver.
3. Configure the ELK M1 XEP IP address and port in the Setup Dialog.
4. Enable optional pulse telemetry if a Hall sensor is installed.
5. Enable optional scope safety interlock if safety sensors are installed.
6. Verify ElkRP text and automation rules are configured properly.
7. Connect the driver from NINA or another ASCOM client.

Connection methods

Connection Methods

The m1OASYS driver supports two communication methods:

TCP/IP (Elk XEP)
   Connects through the Elk XEP Ethernet interface.

Serial
   Connects directly to the Elk M1 RS-232 port at 115200 baud.

Both connection methods use the native Elk ASCII protocol and provide identical functionality including:

• Roof control
• Pulse telemetry
• Calibration
• Scope safety
• Watchdog protection
• Live status monitoring

Calibration

Calibration

The driver supports automatic roof calibration using Hall-effect pulse counting. During calibration, the roof opens fully while the driver records the pulse count. The learned value is saved automatically and used for percent-open calculations.

Telemetry

- Roof state
- Percent open
- Pulse count
- Fault state
- Reconnect status
- Watchdog events
- Calibration status

Roof Position Tracking and Calibration

Both the RRCI and m1OASYS roof controllers use a combination of hall-effect pulse counting and physical open/closed limit sensors to determine roof position and movement status.

Hall Pulse Counting

As the roof moves, a hall-effect sensor generates pulses that are counted by the controller. During calibration, the roof is moved from the fully closed position to the fully open position and the total pulse count is recorded as the calibrated travel distance.

This calibrated value is stored and used to estimate roof position as a percentage:

- 0% = Fully Closed
- 100% = Fully Open
- Intermediate values are calculated from the current pulse count relative to the calibrated full-travel pulse count.

Open and Closed Limit Sensors

Physical limit sensors provide positive confirmation when the roof reaches either end of travel.

When the Open limit sensor is active, the driver considers the roof fully open regardless of pulse count.

When the Closed limit sensor is active, the driver considers the roof fully closed regardless of pulse count.

These sensors serve as the authoritative source for endpoint position and provide protection against lost pulse counts, controller restarts, or communication interruptions.

Reconnect and Recovery Behavior

If the controller reconnects after a restart and pulse information is unavailable, the driver uses the state of the limit sensors to restore roof position.

For example:

- Open sensor active → Roof position restored to 100% open.
- Closed sensor active → Roof position restored to 0% open.

This ensures that roof status remains accurate even if the controller's internal pulse counter has been reset.

m1OASYS Driver

The m1OASYS driver relies primarily on controller-reported roof state and limit sensor status. Hall pulse telemetry is used to provide position estimates, progress indication, and calibration information, but the physical limit sensors remain the authoritative indication of whether the roof is fully open or fully closed.

Summary

- Hall pulses provide continuous position tracking during movement.
- Open and Closed limit sensors provide authoritative endpoint verification.
- Calibration establishes the full-travel pulse count.
- Reconnect recovery uses limit sensor status to restore accurate roof position if pulse information is unavailable.

This combination provides both accurate position tracking and reliable operation during controller restarts, communication interruptions, or recovery from fault conditions.

Using USR TCP/IP to Serial

Using a USR-TCP232-302 TCP/IP Serial Device Server with an ELK M1

m1OASYS supports communication with an ELK M1 through a USR-TCP232-302 serial-to-Ethernet device server. This allows the ELK panel to be accessed over a local network using either a TCP/IP connection or a Virtual COM Port (VCOM).

Recommended USR Configuration

Configure the USR-TCP232-302 as follows:

- Work Mode: TCP Server
- Local Port: 2102 (or another unused port)
- Baud Rate: 9600
- Data Bits: 8
- Parity: None
- Stop Bits: 1
- RFC2217: Disabled
- Short Connection: Disabled
- Link: Enabled

- Packing Time: 0
- Packing Length: 0

After saving the configuration, reboot the USR if necessary.

Virtual COM Port Configuration

If using the USR VCOM utility:

- Configure the VCOM connection as a TCP Client.
- Enter the IP address of the USR device.
- Use the same TCP port configured in the USR (2102 by default).
- Create a virtual COM port and verify that the VCOM status indicates "Connected."

Important Cable Requirement

The ELK M1 Port 0 serial connector and the USR-TCP232-302 serial connector are both wired as DCE devices. A standard straight-through serial cable will not allow communication between them.

A null modem adapter or null modem cable must be used between the ELK and the USR.

Minimum required wiring:

- Pin 2 ↔ Pin 3
- Pin 3 ↔ Pin 2
- Pin 5 ↔ Pin 5

If communication problems are encountered, verify the null modem adapter with a continuity meter. Some adapters are incorrectly wired or defective.

Verifying Communication

Before troubleshooting m1OASYS, verify communication independently:

1. Connect the ELK M1 to the USR using a null modem adapter.
2. Start the USR VCOM utility and connect to the device.
3. Open a terminal program such as PuTTY on the assigned virtual COM port.
4. Observe incoming ELK messages.

Typical unsolicited ELK messages include:

- XK messages (time and date updates)
- KC messages (counter/status updates)

Example:

16XK29141041006261100070

19KC01000000000333333300FF

If these messages are visible in the terminal program, the ELK, cabling, USR device, and VCOM configuration are operating correctly.

Connection Verification in m1OASYS

m1OASYS does not consider the ELK connected merely because a COM port or TCP socket opens successfully. The driver waits for valid ELK traffic before reporting a successful connection.

A connection is considered established when a valid ELK message such as VN, XK, KC, XX, or D6Z is received from the panel.

This prevents false-positive connections caused by incorrect COM port selections, disconnected cables, or inactive serial servers.

Safety Features

Safety Features

Movement Timeout Watchdog

Detects stalled motion, failed relays, missing limit transitions, or roof jams. Automatically aborts motion and reports a fault state.

Pulse Watchdog Protection

When pulse telemetry is enabled, the driver monitors Hall pulses during roof movement. If motion pulses stop unexpectedly, the driver enters a fault state.

Scope Safety Interlock

Optional hardware-based scope-safe interlocks can be enabled through ELK rules. When enabled, the driver blocks roof movement unless the ELK reports the telescope is in a safe position.